Physics for EEE 2022 scheme

Question Bank

Mudule-5 (Semiconductors & Devices)

1. Mention the expression for electron concentration and hole concentration in an intrinsic semiconductor and explain the terms.
2. Explain Fermi level in an intrinsic semiconductor. Give its significance.
3. Briefly explain variation of Fermi level in P-type & N-type Semiconductors.
4. Derive the relation between Fermi energy and energy gap for an intrinsic semiconductor (or Show that $E_F = \frac{E_g}{2}$ for intrinsic semiconductor).
5. State and explain law of mass action.
6. Derive an expression for electrical conductivity of a semiconductor.
7. What is Hall Effect? Obtain an expression for the Hall voltage in terms of Hall coefficient. Mention few applications of Hall Effect.
8. Explain the construction and working of photodiode? Discuss the power responsivity in a photodiode. Mention few uses of photodiode.
9. Explain the construction and working of photo transistor and also mention any two applications. Mention few uses of phototransistor.
10. Explain how the resistivity of a semiconductor is determined using four probe method? Mention any two applications of four probe method.